Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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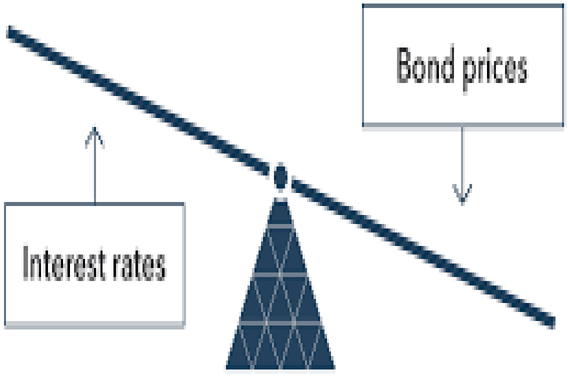
Interest Rates and Valuation

(Chapter 3)



Interest rates variation over time

Chapter Preview

Interest rates are among the most closely watched variables in the economy.

This lecture: Lots of definitions and (some) math.

IMPORTANT:

LEARN by UNDERSTANDING,

DO NOT LEARN BY HEART!

Topics

1. Present value
2. Yield to maturity
3. Zero coupon bonds
4. Coupon bonds
5. Perpetuity
6. Nominal Vs real interest rate
7. Rate of return and interest rate risk
8. Duration

Present Value: Introduction

Suppose we have 100€ and 50$. What is our total wealth?

$$100€ + 50\$ = ??$$

We need an <u>exchange rate</u> to express everything in a common unit (base currency)

$$100€ + 50\$ \times (0.84\ €/\$) = 142€$$

$$100€ \times (1.19\ \$/€) + 50\$ = 169\$$$

Now consider 1$ today (t=0) and 1$ tomorrow (t=1):

$$1\$_0 + 1\$_1 = \mathbf{??}$$

No change of currency, but change of **time**!

Present Value: Introduction

- ✓ Consider a 1$ today (t=0) and a 1$ tomorrow (t=1)
- ✓ 1$ today (time t=0: $\$_0$) is **NOT** equal to 1$ tomorrow (time t=1: $\$_1$).

Why?

- ✓ Presumably 1$ today is <u>better</u> than a 1$ tomorrow, because:
  a. the future is uncertain
  b. people are impatient

- ✓ To find the value of $\$_0$ corresponding $\$_1$ we need a conversion rate (or a sort of "*exchange* rate") between today and tomorrow ($\$_0$ /$\$_1$):

discount factor

Present Value: Introduction

Discount factor

✓ Invest 1 today, get $(1+i)$, with $i>0$ tomorrow. Hence:

$$\$_1 = (1+i)\$_0 \text{ , with } i>0$$

$\$_1 = (1+i)\$_0$

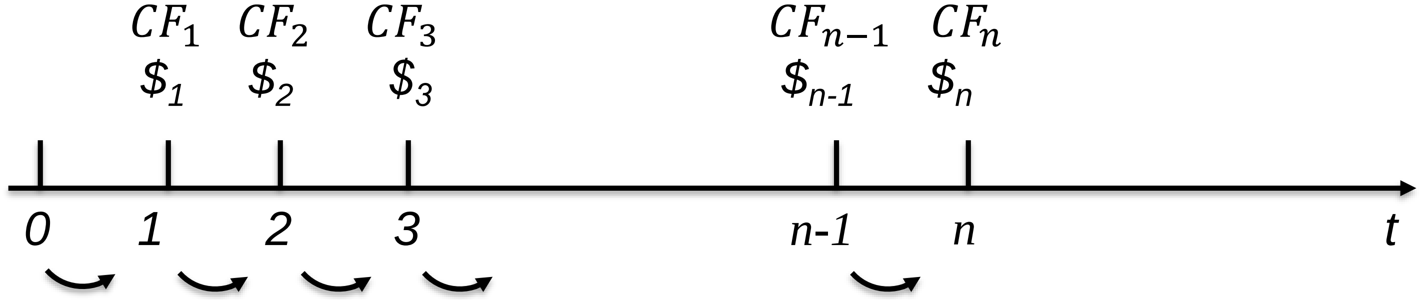
$\$_2 = (1+i)\$_1 = (1+i)(1+i)\$_0 = (1+i)^2\$_0$

$\$_3 = (1+i)\$_2 = (1+i)(1+i)(1+i)\$_0 = (1+i)^3\$_0$

…

$\$_n = (1+i)\$_{n-1} = \ldots = (1+i)^n\$_0$

At each period t have a cash flow ($CF_t = \$_t$) that is fully reinvested with interest earned in the previous period:



Present Value: Introduction

Discount factor

- ✓ the *discount factor* is a function of the **interest rate** (i) between today and tomorrow
- ✓ it's called *discount* because you are *discounting* the future
- ✓ can see the discount factor in this way: $\$_1 = (1+i)\$_0$, with $i>0$

$$\frac{\$_0}{\$_1} = \boxed{\frac{1}{1+i} = \delta} \quad \text{(1 period)}$$

$$\frac{\$_0}{\$_2} = \frac{\$_0}{\$_1} \cdot \frac{\$_1}{\$_2} = \frac{1}{1+i} \cdot \frac{\$_1}{\$_2} = \frac{1}{1+i} \cdot \frac{1}{1+i} = \boxed{\frac{1}{(1+i)^2} = \delta^2} \quad \text{(2 periods)}$$

$$\frac{\$_0}{\$_3} = \frac{\$_0}{\$_1} \cdot \frac{\$_1}{\$_2} \cdot \frac{\$_2}{\$_3} = \frac{1}{(1+i)^2} \cdot \frac{\$_2}{\$_3} = \frac{1}{(1+i)^2} \cdot \frac{1}{1+i} = \boxed{\frac{1}{(1+i)^3} = \delta^3} \quad \text{(3 periods)}$$

…

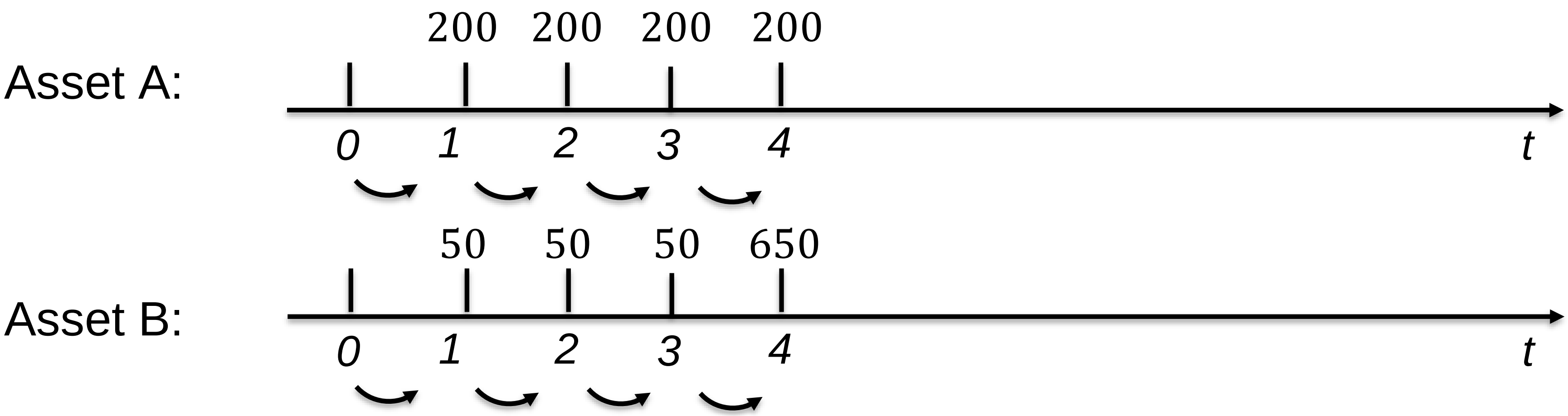
…

$$\frac{\$_0}{\$_n} = \frac{\$_0}{\$_1} \cdot \ldots \cdot \frac{\$_{n-1}}{\$_n} = \frac{1}{(1+i)^{n-1}} \cdot \frac{\$_{n-1}}{\$_n} = \frac{1}{(1+i)^{n-1}} \cdot \frac{1}{1+i} = \boxed{\frac{1}{(1+i)^n} = \delta^n} \quad (n \text{ periods})$$

Present Value: Introduction

PV useful to compare various financial instruments

✓ EX: consider 2 financial instruments (4 ex.: bonds, loans, etc…): A and B



The interest rate is $i = 0.1$ $(= 10\%)$. Then:

✓ $PV_{A,TOT} = \frac{200}{(1.1)} + \frac{200}{(1.1)^2} + \frac{200}{(1.1)^3} + \frac{200}{(1.1)^4} = 634 \neq (200 + 200 + 200 + 200)$

✓ $PV_{B,TOT} = \frac{50}{(1.1)} + \frac{50}{(1.1)^2} + \frac{50}{(1.1)^3} + \frac{650}{(1.1)^4} = 568.3 \neq (50 + 50 + 50 + 650)$

✓ So at *t=0*, asset A is more valueable than asset B, even though sum of cash flows is the same (800 in both cases)! *Why?*

Present Value: Introduction

Debt instruments differ by:

✓ streams of cash payments (**cash flows**)

… and …

✓ timing

The **evaluation** of amounts at different points in time is called ***Present Value (PV)*** *analysis*

All present value calculations are of this kind:

$$PV = f\left(\frac{Stream\ of\ Future\ Cash\ Flows}{1+i}\right)$$

PV computes the value at time 0 (=present) of future cash flows

Present Value: Introduction

In previous formula we call i the **yield-to-maturity (YTM)*** or simply **interest rate**

IF

the price of the security (*P*) is equal to the Present Value (*PV*) of its future cash flows

$$\boxed{\text{if } P = PV \;\rightarrow\; i = YTM}$$

Definition: *YTM is the interest rate that equates the PV of the cash flows with the value (price) of the debt instrument today*

Note that: $\boldsymbol{i} \neq$ **RATE OF RETURN**

We will see examples where $\boldsymbol{i}$>0 but Return<0

*: think of YTM as the int. rate that allows to make PV of an asset until maturity

Present Value: Introduction

In most finance applications:

$$P = PV(\textit{cash flows})$$

where P is the Security's Price

P=PV is true under certain assumption on investors' rationality and **efficiency** of financial markets

We will see this later in the course …

… for now, just assume it: financial markets are efficient

Present Value: Applications

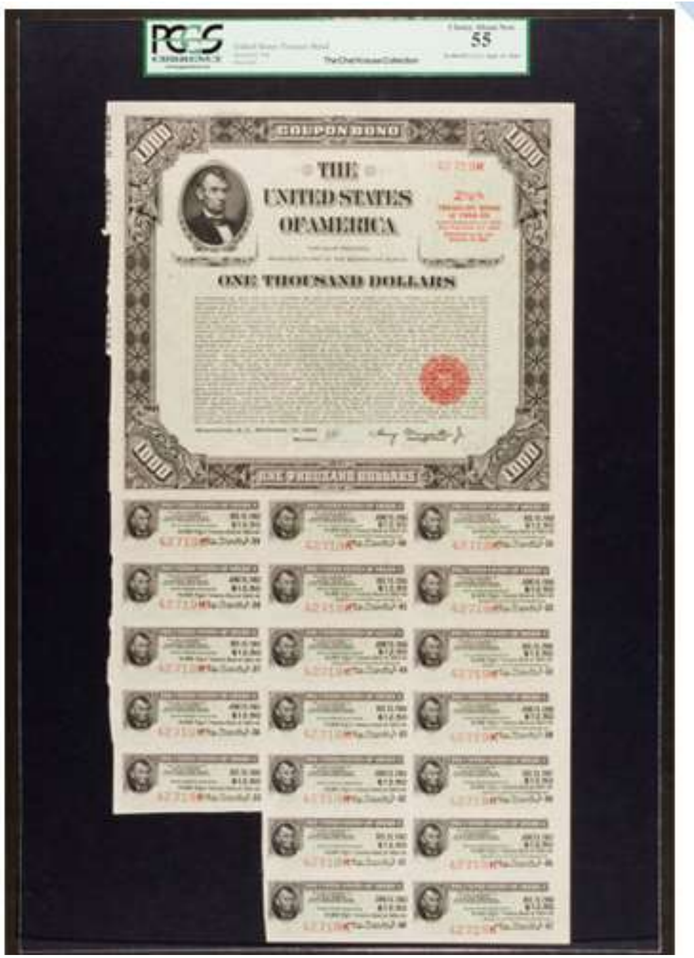
- ✓ There are two basic types of debt instruments which incorporate present value concepts:
  - **LOANS** (for ex.: mortgages)
  - **BONDS** (for ex.: Treasury or corporate bonds)

- ✓ We'll only look at bonds in this class (examples with loans in the book). In particular:

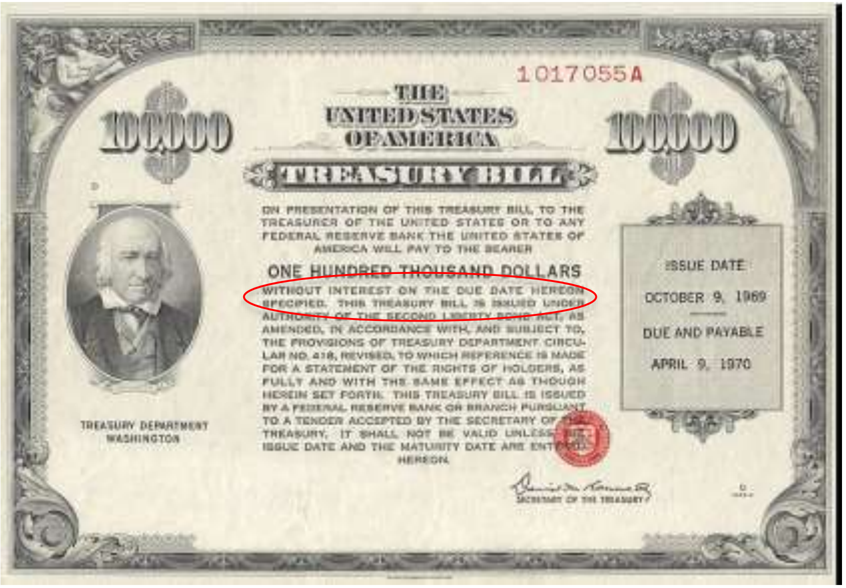
1. **Zero Coupon Bond (or Discount Bond)**
   - No coupon

2. **Coupon Bond**
   - With coupon

Coupon and Zero-coupon bonds



[Coupon bond from US Treasury]



[Zero Coupon bond from US Treasury]

Discount or Zero-Coupon Bond (ZCB)

A ZCB (or call ir **discount bond**):

- ✓ does *not* give *any* interest (coupon) payments before maturity (=zero-coupon*)* *…*
- ✓ … but it only pays the face value (*FV*) to the holder of the bond at the end (=maturity)

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- ✓ … but it only pays the face value (*FV*) to the holder of the bond at the end (=maturity)
- ✓ It normally sells at a price *below* the face value (at discount)
- ✓ Ex: pay $0.95 today, get $1 at maturity. Earning=$0.05. YTM?

Discount or Zero-Coupon Bond (ZCB)

A ZCB (or call ir **discount bond**):

- ✓ does *not* give *any* interest (coupon) payments before maturity (=zero-coupon*)* …
- ✓ … but it only pays the face value (*FV*) to the holder of the bond at the end (=maturity)
- ✓ It normally sells at a price *below* the face value (at discount)
- ✓ Ex: Calculate the **YTM** for a zero-coupon that pays \$0.95 today, get \$1 at maturity (assume 1 year maturity). Intuitively:

$$i = \frac{1-0.95}{0.95} = 0.053 \text{ (read: 5.3\%)}$$

So, formula is: $i = \frac{FV-P}{P}$

- ✓ Government bills are typically ZCBs
  - The US T-bills are the most widely traded discount bonds

Discount or zero-coupon Bond

- ✓ Price of a discount bond with **1** year maturity:

$$i = \frac{FV - P}{P} \Longrightarrow i = \frac{FV}{P} - 1 \Longrightarrow 1 + i = \frac{FV}{P} \Longrightarrow P = \frac{FV}{1 + i}$$

Discount or zero-coupon Bond

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$$i = \frac{FV - P}{P} \Longrightarrow i = \frac{FV}{P} - 1 \Longrightarrow 1 + i = \frac{FV}{P} \Longrightarrow P = \frac{FV}{1+i}$$

- ✓ Price of a discount bond with ***n*** years maturity instead:

$$P = \frac{FV}{(1+i)^n} \Rightarrow i = \sqrt[n]{\frac{FV}{P}} - 1$$

  - Why? Recall $\frac{\$_0}{\$_n} = \frac{1}{(1+i)^{\boldsymbol{n}}} \Rightarrow (1+i)^{\boldsymbol{n}} = \frac{\$_n}{\$_0} \Rightarrow 1 + i = \left(\frac{\$_n}{\$_0}\right)^{\frac{1}{n}} \Rightarrow i = \left(\frac{\$_n}{\$_0}\right)^{\frac{1}{n}}$-1

Discount or zero-coupon Bond

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  - *Example*: what is the YTM for a discount bond with 5 years maturity with a face value of $1,000, selling at $750?

$$n = 5;\ FV = \$1{,}000;\ P = \$750$$

$$i = \sqrt[5]{\frac{\$1{,}000}{\$750}} - 1 \Rightarrow i = 0.0592 \text{ (read 5.92\%)}$$

Coupon Bonds

A **coupon bond**

- ✓ is a bond that makes a fixed payment at specific dates (<u>coupons</u>) plus a final amount (<u>face value or par value</u>) at maturity

- ✓ The price today of a coupon bond is the PV of its future cash flows

- ✓ Consider ***2*** periods:

$$P = \frac{C}{(1+i)} + \frac{C}{(1+i)^{2}} + \frac{\boldsymbol{FV}}{(\mathbf{1}+\boldsymbol{i})^{\mathbf{2}}}$$

  - where C is the coupon payment, FV is the face value, n the years to maturity, *i* is the **YTM** (C, i fixed over time)

Coupon Bonds

- ✓ Consider **many** periods. For ex., a 10% coupon bond with a face value of $1,000 and 10 years to maturity will have a cash flow of $100 each year plus a payment of $1,000 at the end:

$$P = \frac{C}{(1+i)} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \cdots + \frac{\boldsymbol{C}}{(\boldsymbol{1}+\boldsymbol{i})^{\boldsymbol{10}}} + \frac{\boldsymbol{FV}}{(\boldsymbol{1}+\boldsymbol{i})^{\boldsymbol{10}}}$$

$$P = \frac{\$100}{(1+i)} + \frac{\$100}{(1+i)^2} + \frac{\$100}{(1+i)^3} + \cdots + \frac{\$\boldsymbol{100}}{(\boldsymbol{1}+\boldsymbol{i})^{\boldsymbol{10}}} + \frac{\$\boldsymbol{1,000}}{(\boldsymbol{1}+\boldsymbol{i})^{\boldsymbol{10}}}$$

- ✓ Generic formula of price of CB's:

$$P = \sum_{t=\mathbf{1}}^{n} \left[\frac{C}{(1+i)^t}\right] + \frac{FV}{(1+i)^n}$$

  - Note time convention: Σ starts at $t = \mathbf{1}$ (assume buy coupon bond at $t = 0$, but bond only starts paying the following year) and it ends at maturity n

Coupon Bonds

Useful terminology

- ✓ ***Coupon rate***: the amount it pays every year is expressed as a % of face value:

$$i_{coupon\ rate} = \frac{Coupon}{Face\ Value} = \frac{C}{\boldsymbol{FV}}$$

- ✓ If $P = FV$ we have a **par bond**
- ✓ If $P < FV$ we have a bond **at discount**
- ✓ If $P > FV$ we have a bond **at premium**

CBs Example: P, YTM and coupon rate

Recall the previous example:

$$P = \frac{C}{1+i} + \frac{C}{(1+i)^2} + \cdots + \frac{C}{(1+i)^{10}} + \frac{FV}{(1+i)^{10}}$$

Q: if the YTM=11.75%, what is the price today of a "10% coupon bond" with a face value of $1,000 and 10 years maturity?

CBs Example: P, YTM and coupon rate

Recall the previous example:

$$P = \frac{C}{1+i} + \frac{C}{(1+i)^2} + \cdots + \frac{C}{(1+i)^{10}} + \frac{FV}{(1+i)^{10}}$$

Q: if the YTM=11.75%, what is the price today of a “10% coupon bond” with a face value of $1,000 and 10 years maturity?

A: First recover coupon value C:

$$i_{coupon\,rate} = \frac{C}{FV} \Longrightarrow C = i_{cr} \times FV \Longrightarrow C = 0.1 \times \$1{,}000 = \$100$$

Then, calculate price of CB using PV of cash flows:

$$P = \frac{\$100}{1.1175} + \frac{\$100}{(1.1175)^2} + \cdots + \frac{\$100}{(1.1175)^{10}} + \frac{\$1{,}000}{(1.1175)^{10}} = \$900.1$$

or apply generic formula: $P = \sum_{t=\mathbf{1}}^{n} \left[\frac{C}{(1+i)^t}\right] + \frac{FV}{(1+i)^n}$

$$\sum_{t=1}^{10} \left[\frac{\$100}{(1+0.1175)^t}\right] + \frac{\$1{,}000}{(1+0.1175)^{10}} = \$900.1$$

CBs Example: P, YTM and coupon rate

A 3-year corporate bond has a face value of $1,000 and a 7% coupon rate; the current market interest rate is 6%.

$[n = 3;\ FV = \$1{,}000;\ i_{cr} = 0.07; i = 0.06]$

Q: What should be the bond's price?

CBs Example: P, YTM and coupon rate

A 3-year corporate bond has a face value of $1,000 and a 7% coupon rate; the current market interest rate is 6%.

$[n = 3;\ FV = \$1{,}000;\ i_{cr} = 0.07; i = 0.06]$

Q: What should be the bond's price?

R: Recall:

- $i_{cr} = \frac{C}{FV}; C = i_{cr} \times FV = 0.07 \times \$1{,}000; C = \$70$

$$P = \frac{\$70}{1.06} + \frac{\$70}{(1.06)^2} + \frac{\$70}{(1.06)^3} + \frac{\$1{,}000}{(1.06)^3} = \$1{,}026.73$$

- So $P = \$1{,}026.73$

Special case of CB: Perpetuity

A special case of a coupon bond is a **perpetuity** (or **consol**)

✓ It is a coupon bond providing a periodic coupon ***forever*** (=with no maturity date)

✓ It is convenient in finance because it is easy to calculate its price (*i.e.* its present value)

$$PV = \sum_{t=1}^{\infty} \frac{C}{(1+i)^t} = C \sum_{t=1}^{\infty} \frac{1}{(1+i)^t} = \frac{C}{i} \Longrightarrow \boxed{\boldsymbol{PV = \frac{C}{i}}}$$

- can see *immediately* **inverse relation btw P and i**

✓ Why? If you like math here is why. It is a geometric series:

$$1 + x + x^2 + +x^3 \ldots = \sum_{t=0}^{\infty} x^t = 1 + \sum_{t=1}^{\infty} x^t = \frac{1}{1-x} \quad \text{for } |x| < 1$$

Set $x = \frac{1}{1+i}$, then:

$$\sum_{t=1}^{\infty} \frac{1}{(1+i)^t} = \sum_{t=0}^{\infty} \frac{1}{(1+i)^t} - 1 = \frac{1}{1-\left(\frac{1}{1+i}\right)} - 1 = \frac{1+i}{i} - 1 = \frac{1}{i}$$

Do perpetuities exist?

- ✓ UK government issued perpetual war bonds in WWI-WWII:
  - "*If you cannot fight, you can help your country by investing all you can in 5 per cent Exchequer Bonds ... Unlike the soldier, the investor runs no risk.*"
- ✓ These war bonds were redeemed by the UK government in October 2014
- ✓ Other remaining consols dating even further back (1850s) were redeemed in July 2015
- ✓ Nowadays perpetuities are issued by banks for regulatory purposes

Current yield

- ✓ The perpetuity allows us to introduce the concept of **current yield**:

$$i_{current} = \frac{C}{P}$$

- ✓ The current yield is a useful approximation to the YTM for <u>long-term bonds</u>, with <u>price near par</u>.

- ✓ Why? Because CF very distant in time have a low PV, so a 30-year bond is pretty much like a perpetuity!

Example:

i. a perpetuity bond with P=\$2,000 and C=\$100 has $i_{current} = \frac{\$100}{\$2{,}000} = 0.05$ (=<u>5</u>%)

ii. a coupon bond with same price and coupon, with **30** years of maturity (and a FV=\$2,500) has………………………………………… *i* = <u>5.354</u>%

iii. a coupon bond similar to above with **50** years maturity: *i* = <u>5.115</u>%;

iv. a coupon bond similar to above with **100** ys maturity: *i* = <u>5.010</u>%

v. … and so on

Current yield and Coupon Rate

Notice the distinction between

- ✓ The ***current yield*** (Coupon divided by the Price):

$$i_{current} = \frac{C}{\boldsymbol{P}}$$

- ✓ The ***coupon rate*** (Coupon divided by the Face Value)

$$i_{coupon} = \frac{C}{\boldsymbol{FV}}$$

- ✓ of course for a *par-bond* the two coincide (bcs $P = FV$).

Learn these definitions well so avoid confusion later!

YTM and Price

So what is more important, the YTM or the price?

- ✓ <u>mathematically</u>, it does not matter: given one, the other one is determined via the PV formula
- ✓ but <u>economically</u> we think of the YTM <u>determining</u> the price, not viceversa.
- ✓ Why? Think about the YTM as the **“fair” return** an investor **requires** considering the risk
- ✓ Then the investors will price the bonds so that in equilibrium they get this fair return

Relationships between Price, YTM and Coupon rate

Table 3.1 Yields to Maturity on a 10% Coupon rate bond, maturing in 10 years (Face Value = $1,000)

Price of Bond ($)	Yield to Maturity (%)
1,200	7.13
1,100	8.48
1,000	10.00
900	11.75
800	13.81

Three interesting facts in Table 3.1:

1. **When bond is at par** (P=FV)**, YTM equals coupon rate**
2. **Price and yield are negatively related (VERY IMPORTANT!)**
3. **Yield greater than coupon rate when bond price is below par value**

Relationships between Price, YTM and Coupon rate

1) When bond is at par, yield equals coupon rate.

- math: if $P = FV$, then $i_{current}\left(=\frac{C}{P}\right)$ and $i_{coupon}\left(=\frac{C}{FV}\right)$ coincide
- intuition: a *par bond* is like a bank account, if you put down \$1,000 (*P*) today (*t=0*) and you cash in the interest payment every year (*C*=\$100), you are left with \$1,000 (FV) at maturity (say *t=10*) …
- … similar to coupon bond with i = 10% $\left(\frac{C}{FV} = \frac{\$100}{\$1{,}000} = 0.1\right)$ and 10 years maturity

2) Price and yield are negatively related

- **If *i* increases, the PV of any given cash flow is lower; hence, the price of the bond must be lower**
- a bit more obvious from perpetuity and zero-coupon bond formulas, but it emerges also from coupon bonds

Relationships between
Price, YTM and Coupon rate

3) The YTM is greater than the coupon rate when bond price is below par value

- According to (1): $if\ P = FV \Rightarrow i = \frac{C}{FV}$
- According to (2): $if\ i \uparrow then\ P \downarrow$
- Putting (1) and (2) together, it must be that:

$$if\ i \uparrow \Rightarrow YTM > i_{cr}\left(= \frac{C}{FV}\right)\ and\ since\ P \downarrow \Rightarrow\ P < FV$$

What does this mean?

- ✓ If the <u>required yield on the bond is higher than what the bond actually pays</u> (that is, the coupon rate), then it needs **to sell at discount to attract investors** (=price is too high, not attractive!)

- ✓ In other words, because investors can make a larger return the financial market (=other securities), they need an extra incentive to invest in the bond

Real and Nominal Interest Rates

- ✓ The **real interest rate** (i_r) is the nominal interest rate (i) adjusted for **expected inflation** (expected changes in the price level).
  - From the Fisher equation:

$$i_r = i - \pi^e$$

- ✓ Technically, this definition of i_r is the ***ex-ante*** **real rate of interest** because it is adjusted for the ***expected*** level of inflation, π^e

- ✓ We can calculate the ***ex-post*** real rate based on the observed level of inflation (π)

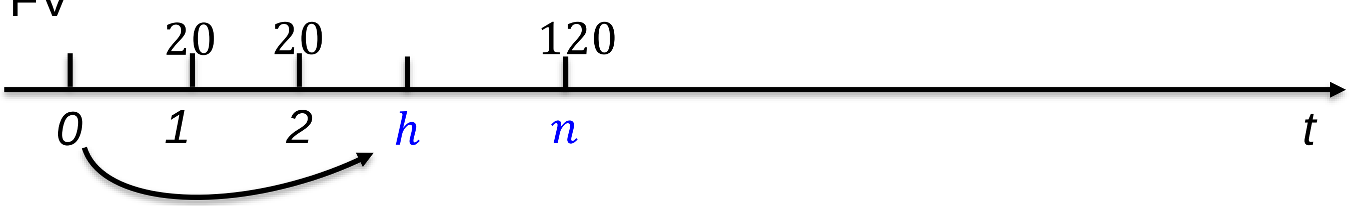
Real and Nominal Interest Rates

- ✓ When the real rate (i_r) is low:
  - there are **greater** incentives to **borrow** (money is less costly)
  - and **less** incentives to **lend** (money is not very profitable)
- ✓ In response to the Financial Crisis (2008), Fed & ECB turned $\underline{i = 0}$, so $\underline{i_r < 0}$ (inflation was low but still positive) to stimulate aggregate spending (consumption and investment)
- ✓ *Case of Deflation* (π^e<*0*):
  - the real rate will always be positive: consider case of lowest i ($i = 0$):
    - $i_r = -\pi^e > 0$ bcs $\pi^e < 0$
  - Deflation can be a big problem:
    - expectations of deflation lead to higher real rates and may cause deflationary *spirals*: The Economist, 07 Jan 2015
    - but some evidence says it is not so bad: CEPR, Jan 2022

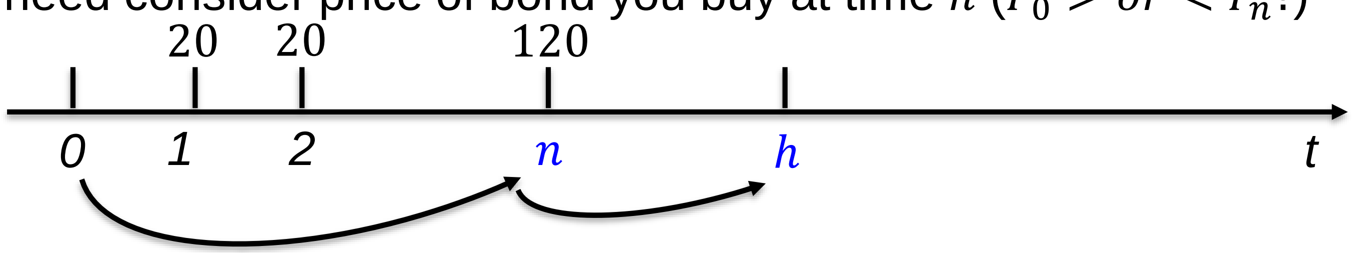
Do not hold bond until maturity

Let h = holding period and n = maturity

- ✓ *So far mostly considered $h = n$ (hold bond until maturity)*
- ✓ *Now consider 2 cases where* holding period $\neq$ maturity
- ✓ Case **1**: you sell the bond at time t **before** maturity (n)
  - holding period shorter than maturity ($\boldsymbol{h < n}$)
  - need consider price of sale at time h (P_h may be different from P_0), bcs do not get FV



- ✓ Case **2**: you sell bond at time t **after** maturity (n)
  - holding period shorter than maturity ($\boldsymbol{h > n}$)
  - need consider price of bond you buy at time n ($P_0 > or < P_n$?)



Interest Rate ≠ Rate of Return

Rate of return

- ✓ Suppose we buy a coupon bond, we hold it for only <u>one period</u> and we <u>don't keep it up to maturity</u>, but we **sell it before maturity (**$\boldsymbol{h < n}$**)**
  - with sale *before* maturity, <u>price</u> may <u>change</u> (we don't wait to get FV)
  - so, we need to consider the <u>**capital gain**</u> (say: if cap. gain is negative, we may have a loss!)

- ✓ The <u>*rate of return*</u> on our investment is all the payments received during the holding period, divided by the initial price:

$$Return = \frac{C+P_{t+1}-P_t}{P_t} = \frac{C}{P_t} + \frac{P_{t+1}-P_t}{P_t} = i_c + g$$

where:

$i_c = \frac{C}{P_t}$ (current yield)

and

$g = \frac{P_{t+1}-P_t}{P_t}$ (capital gain)

Example: Interest Rate ≠ Rate of Return

Assume you paid $1,000 for a 10-year coupon bond with a face value of $1,000, and a coupon rate of 10%.

One year after the purchase, you sold the bond for:

a. $1,200

b. $ 800

Q: What is your *rate of return*? What is the (initial) YTM?

Example: Interest Rate ≠ Rate of Return

Assume you paid $1,000 for a 10-year coupon bond with a face value of $1,000, and a coupon rate of 10%.

One year after the purchase, you sold the bond for:

a. $1,200

b. $ 800

Q: What is your *rate of return*? What is the (initial) YTM?

A: $C = i_{cr} \times FV = 0.1 \times \$1{,}000 = \$100$

(a): $Return = \frac{\$100+(1{,}200-\$1{,}000)}{\$1{,}000} = \frac{\$300}{\$1{,}000} = \quad 0.3 \text{ (read: } 30\%)$

(b): $Return = \frac{\$100+(\$800-\$1{,}000)}{\$1{,}000} = -\frac{\$100}{\$1{,}000} = -0.1 \text{ (read: } -10\%)$

$YTM = i_{cr}$ bcs bond is at par; YTM=10%

So returns differ between (a) and (b), even though same $YTM = 10\%$

Example: Interest Rate ≠ Rate of Return

What happened after one year?

- ✓ by the time you wanted to sell the bond, its yield
  - went **down** (=price went up) in case a
  - went **up** (=price went down) in case b

- ✓ Case a: new investors are willing to pay a higher price for a bond whose coupon rate (10%) is higher than current interest rates at prevailing market conditions
  - reverse in case b

- ✓ Let's see in the next table what happens to bonds' returns if **the interest rate increases**, depending on the bond *different maturities*

Key Facts about the Relationship between Rates and Returns

Table 3.2 One-Year Returns on 10% Coupon Par Bonds (FV=$1,000)

When interest rate rises <u>from 10% to 20%</u>

(1) Years to Maturity When Bond Is Purchased	(2) Initial Current Yield (%)	(3) Initial Price ($)	(4) Price Next Year* ($)	(5) Rate of Capital Gain (%)	(6) Rate of Return (2 + 5) (%)
30	10	1,000	503	–49.7	–39.7
20	10	1,000	516	–48.4	–38.4
10	10	1,000	597	–40.3	–30.3
5	10	1,000	741	–25.9	–15.9
2	10	1,000	917	–8.3	+ 1.7
1	10	1,000	1,000	0.0	+10.0

*Calculated with a financial calculator using Equation 3.

Recall the coupon bond formula and return formula:

$$P = \sum_{t=1}^{n} \frac{C}{(1+i)^t} + \frac{FV}{(1+i)^n} \qquad Return = \frac{C+P_{t+1}-P_t}{P_t} = i_c + g$$

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Summary on relationship between rates and returns

Key findings from Table 3.2

1. For bonds with holding period < maturity ($h < n$), $i \uparrow \Rightarrow P \downarrow$, implying capital loss

2. The longer the maturity, the greater the ΔP (and also greater of ΔR) for any Δi

3. The only bond whose return = yield is the one with maturity = holding period (obvious: at maturity, no risk of capital gain/loss!)

4. Even if a bond has a high initial int. rate, the return can turn negative if $i \uparrow$

Interest Rate Risk

- ✓ For the same $\Delta i \uparrow$, ΔP is more negative for L-T bonds … so
- ✓ prices are **more "volatile" for long-term (L-T)** bonds
  - ○ and thus, these bonds are riskier than short term bonds
- ✓ The risk of losing money ($\Delta P < 0$) if interest rates change is called **Interest Rate Risk (IRR)**
- ✓ IRR comes from the fact that the bonds may be **sold before maturity** ($h < n$)
  - ○ in this case you don't know what the interest rate (and **hence the price**) will be at the time you wan to sell it ($t = h$)
  - ○ it is very important for investors (and banks in particular) to manage IRR, not just from bonds, but also other securities

Interest Rate Risk

- ✓ So have IRR only for bonds whose holding period is shorter than maturity ($h < n$)

- ✓ No IRR for any bond whose maturity equals holding period (*i*=return): obvious! ($h = n$)
  - call this case as **hold-to-maturity (HTM)**

- ✓ Of course, we always have ALSO other risks:
  - ***default*** *risk*
  - ***inflation*** *risk* [that's true of any investment in securities that pays a nominal return (some bonds are indexed with interest that adjusts with inflation rate)]

Reinvestment Risk

- ✓ If **holding period is longer than maturity** $(h > n)$, you don't have IRR, but you have **reinvestment risk**
  - at the moment of reinvesting, you may fail to get the same interest rate as before!
  - coupon payments of a coupon bond are $'s that could/should be re-invested in other bonds. But the **future *i* is uncertain**!

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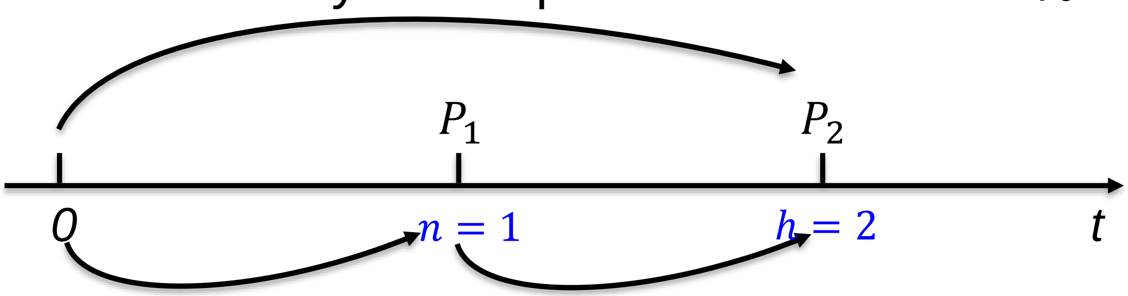
Q: Is it better to invest at time *t* in:

a. a **2**-year coupon bond with $i = 10\%$ with $h = n$

or

b. a **1**-year coupon bond with $i = 10\%$, and re-invest the proceeds at $t + 1$ in another 1-year coupon bond with $i = 10\%$? $(h > n)$?



Uncertainty about i_1 at $t = 0$ determines Reinvestment risk

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A: it depends …

- in case a, $h = 2$ years, while $n = 1$ is 1 year for each bond
- in case b, if interest rate **falls**/**rise** in $t + 1$, you **lose**/**gain**: at the moment of reinvesting, you may fail to get the same interest rate as before!

✓ Bottom line: even very short maturities have risks ...

Duration

Can we **quantify** the **IRR** of a bond?

- the further in the future the *payments* of a bond lie, the more the bond is exposed IRR

- idea: we could use a measure of time of each payment (c.d. "effective maturity") of a bond to quantify IRR

- in practice: to construct this measure of "effective maturity", for each period:
  - 1st: calculate the % of the total PV of a bond is contributed by cash flows in each year payments are due (4 ex.: 1, 2, etc…)
  - 2nd: use these weights to calculate a "weighted" maturity

✓ Duration is the **weighted average of the maturities of the cash payments**

- it's like an <u>effective</u> <u>maturity</u>
- useful because it allows to *quantify* the IRR because …
- … the **longer** the duration, the **higher** the IRR!

Duration

Before we get to the formula, let’s see the intuition

- ✓ We want a measure of IRR
- ✓ Use the idea that the longer the maturity the higher IRR

Two bonds with same original maturity

do **not** necessarily have same IRR

1: a zero-coupon bond (ZCB) makes all payments at the end
2: a coupon bond (CB) spreads them out in each period

Q : is the IRR (measured by the *effective maturity*) higher or lower for a ZCB with the same maturity of the coupon bond?

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Q : is the IRR (measured by the *effective maturity*) higher or lower for a ZCB with the same maturity of the coupon bond?

A: **higher** for ZCB, because ZCB makes all payments at the end, while CB makes payments earlier than ZCB

Formula for Duration

$$DUR = \sum_{t=1}^{n} t \frac{PV_t}{PV_{TOT}} = \sum_{t=1}^{n} t \cdot \left[\frac{\frac{CP_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{CP_t}{(1+i)^t}} \right] = \sum_{t=1}^{n} t \cdot \alpha_t$$

t = maturity of cash flow payment (CP_t)

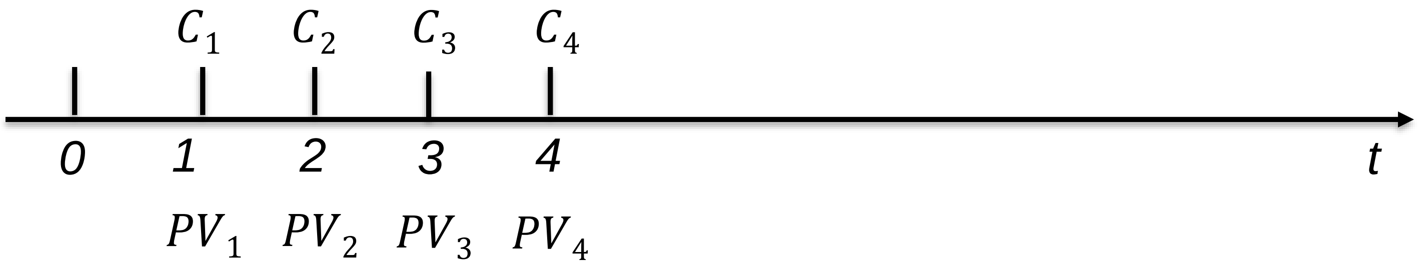
weight

where: $CP_t = C$ for $t < n$; $CP_t = C + FV$ for $t = n$

✓ Formula intuition: the *effective maturity* of coupon bonds is a <u>weighted average</u> of <u>effective maturities</u> on <u>discount bonds</u>

- each coupon <u>payment</u> is like a discount <u>bond</u>
- the weights (α_t) are equal to the proportion of the total value (in terms of PV) represented by each discount bond: $\alpha_t = \frac{\frac{CP_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{CP_t}{(1+i)^t}}$
- **Duration of a ZCB equal to its maturity** (in fact: $\alpha_t = 1$)

Explaining formula for Duration



$$PV_{TOT} = PV_1 + PV_2 + PV_3 + PV_4$$

$\alpha_1 = \frac{PV_1}{PV_{TOT}}$ … and so also $\alpha_2, \alpha_3, \alpha_4$

$$DUR_1 = \alpha_1 \times 1 \;\; \ldots \; DUR_4 = \alpha_4 \times 4$$

$$DUR_t = DUR_1 + DUR_2 + DUR_3 + DUR_4$$

- if the weights are the proportion of the total value (in terms of PV) represented by each discount bond …
- … why is the d**uration of a ZCB equal to its maturity**?
- Because having only one time in which payment takes place (say t), then: $\alpha_t = 1$ by definition

Calculating Duration
i =12%, 5-Year, Coupon Bond

		i =	0.12				
time	1	2	3	4	5		
CF	100	200	0	450	1000		
PV_t	89.3	159.4	0.0	286.0	567.4	PV_{TOT}=	1102.1
α_t	8.1%	14.5%	0.0%	25.9%	51.5%		
DUR_t	0.1	0.3	0.0	1.0	2.6	DUR=	3.98

Decompose PV_{TOT}:

$$PV_{TOT} = \frac{CF_1}{1+i} + \frac{CF_2}{(1+i)^2} + \frac{CF_3}{(1+i)^3} + \frac{CF_4}{(1+i)^4} + \frac{CF_5}{(1+i)^5}$$

$$\alpha_t = \frac{PV_t}{PV_{TOT}}\text{, where: } PV_{TOT} = \sum_{t=1}^{n} PV_t \text{ and } PV_t = \frac{CF_t}{(1+i)^t}$$

$$DUR_t = \alpha_1 \cdot 1\ (= time\ 1) + \alpha_2 \cdot 2 (= time\ 2) + \ ... + \alpha_5 \cdot 5\ (= time\ 5)$$

- ✓ ex: $PV_1 = \frac{100}{1.12} = 89.3; PV_{TOT} = 89.3 + \ ... + 567.4 = 1{,}102.1; \alpha_1 = \frac{89.3}{1{,}102.1} = 0.081$
- ✓ $DUR_t = 0.081 \cdot 1\ + 0.145 \cdot 2\ +\ ...\ + 0.515 \cdot 5\ = 3.98\ years$
- ✓ At home try to replicate it and then change values (*i*, *CF's*, *n*) …

… see what happens to DUR (properties: next slide!)

Calculating Duration

i =10%, 10-Year, 10% Coupon Bond

(1) Year	(2) Cash Payments (Zero-Coupon Bonds) ($)	(3) Present Value (*PV*) of Cash Payments (*i* = 10%) ($)	(4) Weights (% of total *PV* = *PV*/$1,000) (%)	(5) Weighted Maturity (1 × 4)/100 (years)
1	100	90.91	9.091	0.09091
2	100	82.64	8.264	0.16528
3	100	75.13	7.513	0.22539
4	100	68.30	6.830	0.27320
5	100	62.09	6.209	0.31045
6	100	56.44	5.644	0.33864
7	100	51.32	5.132	0.35924
8	100	46.65	4.665	0.37320
9	100	42.41	4.241	0.38169
10	100	38.55	3.855	0.38550
10	1,000	385.54	38.554	3.85500
Total		1,000.00	100.000	6.75850

Duration for a coupon bond always less than actual maturity (see next page)

Properties of Duration

$$DUR = \sum_{t=1}^{n} t \frac{CP_t}{(1+i)^t} \Bigg/ \sum_{t=1}^{n} \frac{CP_t}{(1+i)^t}$$

Properties: DUR↑ if:

1. **Maturity↑** (obvious)

2. $\boldsymbol{i_{cr}}(= \boldsymbol{C/FV})$ ↓ (ZCBs have max duration=actual maturity because they pay all at the end)

3. $\boldsymbol{i}$↓ sensitivity to changes in rates increases as rates fall (because duration is a linear approximation: as int. falls, discount the future less, far distant weights from now are higher)

Recall also:

a. for **ZCB: Duration=Maturity**

b. for **CB: Duration<Maturity**

Duration and Interest-Rate Risk

- ✓ Recall that IRR is given by the price change of a coupon bond if *i* changes
- ✓ With the duration we can <u>approximate</u> (for **small changes** in interest rate) the % price change with:

$$\boxed{\%\Delta P \approx -DUR \times \frac{\Delta i}{1 + i_0}}$$

  - note: $\Delta i = i_1 - i_0$
  - the formula comes from a 1^{st} order Taylor approximation

Ex: if $i \uparrow$ from 10% to 11% for coupon bond with a duration of 6.76 years:

$$\Delta \mathrm{i} = 0.11\text{-}0.1 = 0.01$$

$$\%\Delta P \approx -6.76 \times \frac{0.01}{1 + 0.10} = -0.0615 = -6.15\%$$

Duration and Interest-Rate Risk (cont.)

- ✓ The **greater** the duration of a security, the **greater** the percentage change in the market value of the security (ΔP) for a given change in interest rates (Δi)

- ✓ Therefore, <u>the greater the duration of a security, the greater its IRR</u>

- ✓ The duration is not quite the same as the IRR, but it measures its **intensity**!

Exercise: Duration

A 5% coupon bond has a YTM of 6% and a price of $\$965$.
The interest rate then falls to 5.5%. The duration is 4 years.

Q: What is the new price?

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Q: What is the new price?

A: $P = \$983.2$

Indeed:

$$\Delta \mathrm{i} = 0.055\text{-}0.06 = -0.005$$

$$1+\mathrm{i} = 1+ 0.06 = 1.06$$

$$\%\Delta P \approx -DUR \times \frac{\Delta i}{1+i} = -4 \times \frac{(-0.005)}{1.06} = +0.019$$

$$P_1 = P_0(1 + 0.019) = \$965 \times 1.019 = \$983.2$$

Duration of a Portfolio

Consider not a single security, but a portfolio of securities

Duration is additive (∑)

- ✓ The duration of a *portfolio of securities* is the **weighted-average** of the durations of the individual securities …
- ✓ … with the **weights** equaling the proportion of the portfolio value invested in each security

- ✓ Ex: if have n bonds in the portfolio:
  - $DUR_{portfolio} = DUR_1 \times w_1 + DUR_2 \times w_2 + ... + DUR_n \times w_n$
  - *where w_k is the weight of bond k over total value of bonds in the portfolio*

Exercise: Duration of a Portfolio

- ✓ Assume that you have a portfolio of 2 bonds
  - One has a duration of 3 years, and the other has a duration of 7 years
  - The market values of the bonds are $600 and $900, respectively
  - Both bonds have the face value of $1,000

Q: What is the **duration of your portfolio**?

Exercise: Duration of a Portfolio

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- The market values of the bonds are \$600 and \$900, respectively
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Q: What is the **duration of your portfolio**?

$$DUR_{portfolio} = 3 \times \frac{\$600}{\$600+\$900} + 7 \times \frac{\$900}{\$600+\$900}$$

$$\frac{\$600}{\$600+\$900} = 0.4; \ \frac{\$900}{\$600+\$900} = 0.6$$

$$DUR_{portfolio} = 3 \times 0.4 + 7 \times 0.6 = 1.2 + 4.2 = \ 5.4 \text{ (years)}$$

N.B.1 The weights reflect the values (market prices) of the securities in the portfolio, not their face values

Chapter 3 Summary (1/2)

1. We examined how to price various debt instruments using the Present Value formula
2. We defined the YTM as the interest rate that makes the price of debt (P) equal to its PV
3. We defined the concept of the current yield = C/P
4. We discussed the relation between YTM, P and C
   - Especially that YTM and P are negatively related!
5. We distinguished between real and nominal interest rate

Chapter 3 Summary (2/2)

6. We distinguished between *i* and rate of return and found out that only if we HTM the two coincide

7. Hence, we defined IRR, which makes investment in L-T bonds risky

8. And we've seen how to measure IRR using Duration

Lots of definitions, but these are all **fundamental concepts** for the rest of the semester. Make sure you understand them!